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Symbiosis International (Deemed University), Pune

MINI PROJECT REPORT

SUB: STAIML

The Digital Ethics And News Bias Forensic Analyser

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Abstract

The proliferation of digital news platforms has made media bias detection one of the most pressing challenges in contemporary information science. This report presents a forensic analysis of **60 real-world news headlines** published between February and April 2026, sourced from seven prominent media outlets: TechCrunch, Wired, Reuters, the SEC, McKinsey, the BBC, and the European Commission. The dataset spans three thematic categories -Technology, Finance, and Global Politics -and is enriched with metadata including engagement counts, viral status, impact scores, reach percentages, source trust scores, and reliability classifications.

The study employs a dual-language analytical pipeline, implementing analyses in both **Python (pandas, NumPy, matplotlib, seaborn)** and **R**, to ensure methodological rigor and cross-validation of findings. Core techniques include keyword-based linguistic tone classification, data merging and preprocessing, exploratory data analysis, correlation analysis, viral engagement volatility assessment, and distribution comparison between trusted and unverified content. Key findings reveal that Global Politics headlines generate the highest average engagement (1,269), viral spikes are most concentrated in the European Commission and TechCrunch sources, and content flagged as 'Verify' exhibits a significantly more variable impact distribution compared to 'Trusted' content. These insights contribute to the growing body of evidence-based media literacy frameworks and offer a reproducible methodology for forensic news analysis.

1. Introduction

In the contemporary information landscape, news media plays a central role in shaping public perception, influencing policy, and driving societal narratives. However, the accelerating pace of digital journalism and the fragmentation of news consumption across platforms have given rise to a growing concern: **systemic bias in news reporting**. Whether manifested as selective framing, emotionally charged language, or disproportionate emphasis on certain categories of events, media bias has significant implications for democratic discourse and individual decision-making.

Traditional approaches to media literacy -primarily qualitative and journalist-led -are insufficient for analysing the volume and velocity of modern digital news. The convergence of **statistical methods**, **natural language processing (NLP)**, and **machine learning (ML)** has opened new avenues for rigorous, scalable, and objective bias detection. Forensic media analysis, as a discipline, applies these quantitative tools to interrogate patterns within headline corpora, identifying systemic tendencies in language, tone, source behaviour, and engagement dynamics.

This project addresses the gap between anecdotal bias observation and evidence-based detection by constructing a structured analytical pipeline applied to a curated dataset of 60 news headlines published across seven authoritative sources in early 2026. The specific contributions of this work are:

- Development of a custom Forensic Media Headlines dataset (2026) spanning Technology, Finance, and Global Politics categories.
- Implementation of keyword-based linguistic tone classification to distinguish positive, negative, and neutral framing.
- Statistical profiling of engagement, reach, and virality patterns across sources, categories, and reliability classes.
- Correlation analysis of headline structural properties (length, trust score) with impact and reach outcomes.
- Dual-language implementation of the entire analytical pipeline in Python and R

The remainder of this report is organized as follows: Section 2 reviews related literature on media bias detection and computational journalism; Section 3 describes the proposed dataset, preprocessing pipeline, and analytical methodology; Section 4 presents results with detailed visualizations; Section 5 provides discussion, practical implications, and comparison with prior work; and Section 6 concludes with directions for future research.

2. Related Work / Literature Review

The detection of bias in news media has attracted interdisciplinary attention from journalism scholars, computational linguists, political scientists, and AI researchers. The following review synthesizes key contributions in the field, with particular emphasis on methodologies relevant to this project.

2.1 Political Framing and Computational Approaches

Entman (1993) foundationally defined media framing as the selection and salience of aspects of perceived reality to promote a particular problem definition, causal interpretation, or treatment recommendation. This theoretical grounding inspired subsequent computational efforts. **Recasens et al. (2013)** introduced the Linguistic Bias dataset and trained supervised classifiers to detect epistemically biased language -specifically the use of 'factive' versus 'entailment' verbs in news reporting. Their work demonstrated that lexical choice alone carries significant framing information, achieving F1 scores above 0.70. Their dataset, however, was limited to Wikipedia articles rather than live journalism.

Building on this, **Lim et al. (2020)** applied transformer-based models (BERT fine-tuned on AllSides.com data) to political news bias classification, achieving over 85% accuracy in identifying left, center, and right-leaning articles. The study highlighted the role of source-level metadata -such as outlet political leaning -as a powerful prior in classification tasks. However, their focus was primarily on political axis labeling rather than the broader spectrum of engagement-based and structural bias studied in the present work.

2.2 Engagement Metrics as Proxy for Bias

A growing body of research treats social media engagement metrics as proxies for bias-driven amplification. **Vosoughi, Roy & Aral (2018)** in their landmark Science study found that false news spreads faster, farther, and more broadly than true news on Twitter, with emotional novelty identified as the primary driver. This work is directly relevant to the present study's examination of viral status as a function of content reliability and category. Their methodology -distinguishing 'true' and 'false' cascades through cascade analysis -inspired the operationalization of 'Viral_Status' and 'Reliability_Check' variables in our dataset. A key limitation of their study was its platform specificity (Twitter only), whereas our dataset captures multi-source, multi-domain headlines.

Complementing this, **Horne & Adali (2017)** analysed structural features of news headlines and body text to differentiate real news, fake news, and satire. Their analysis revealed that fake news headlines are shorter, contain fewer technical terms, and use more personal pronouns -findings that motivated the inclusion of 'Headline_Length' as an analytical variable in our dataset. The 'Impact_Score' variable in our study is analogous to their 'article share' metric, though computed as a product of engagement and reach.

2.3 Source Trust and Reliability Classification

The credibility of news sources has been studied extensively in the context of misinformation. **Baly et al. (2018)** proposed a model that predicts the factual reporting level of news media outlets using Wikipedia metadata, web page structure, and Twitter activity, achieving strong predictive performance across multiple reliability dimensions. Their notion of source-level trust inspired the 'Source_Trust_Score' and 'Reliability_Check' columns in our dataset, which stratify sources into 'Trusted' (score 9) and 'Verify' (score 8) categories.

More recent work by **Gruppi et al. (2022)** introduced the NELA-GT benchmark for news reliability, combining 360,000+ articles with publisher-level reliability labels derived from Media Bias/Fact Check and NewsGuard. While their approach is large-scale, it treats reliability as a fixed publisher attribute rather than a per-headline property. Our study extends this by assigning per-headline reliability flags and examining how reliability interacts with engagement and impact at the individual headline level.

2.4 Statistical and Linguistic Methods in Media Analysis

Keyword-based sentiment and tone analysis -the method adopted in this project -has a long tradition in computational text analysis. The LIWC framework (Pennebaker et al., 2001) remains a foundational tool, classifying words into psychological categories including affect, cognition, and social processes. Our tone classification adopts a simplified LIWC-inspired approach, using domain-relevant positive and negative keyword lexicons tailored to the news headlines domain. **Hamborg et al. (2019)** provide a comprehensive survey of automated news bias detection methods, noting that lexicon-based approaches, while limited in contextual depth, offer high interpretability and computational efficiency -properties particularly valuable in pedagogical and forensic settings such as this project.

Summary of Related Literature

Author / Year	Dataset	Methodology	Significance	Accuracy	Limitations
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Recasens et al. (2013)	Wikipedia articles (~1,700 sentences)	Linguistic bias classification; SVM with lexical features	Identified epistemically biased verbs as bias signals	~70% F1	Wikipedia-only; not live journalism
Vosoughi, Roy & Aral (2018)	Twitter cascades; ~126,000 news stories	Cascade analysis; true vs. false news spread	False news spreads faster and farther; novelty drives sharing	Descriptive	Platform-specific (Twitter); binary true/false framing
Horne & Adali (2017)	CREDBANK / BuzzFeed news corpus	Structural text features; SVM + Random Forest	Headline length and vocabulary distinguish real vs. fake news	~74% accuracy	Structural features only; ignores engagement dynamics
Lim et al. (2020)	AllSides.com political articles (~37,000)	BERT fine-tuned for political leaning classification	Source-level metadata improves bias classification	>85% accuracy	Binary political leaning; no engagement analysis
Baly et al. (2018)	Media outlet metadata; Wikipedia + Twitter	Multi-source credibility modeling	Source trust predictable from structural and social signals	~82% accuracy	Source-level granularity; not per-article
Gruppi et al. (2022)	NELA-GT; 360,000+ articles	Publisher reliability labels; multi-label classification	Large-scale benchmark for credibility research	~80% F1	Publisher-level labels; does not vary per headline
Hamborg et al. (2019)	Systematic literature review (150+ papers)	Survey; taxonomy of automated bias detection methods	Comprehensive map of bias detection techniques	Survey	Does not propose a new method; review only
Present Study (2026)	60 custom headlines; 7 sources; 3 categories	Keyword tone, EDA, correlation, volatility, KDE; Python & R	Forensic per-headline analysis with engagement + reliability	Descriptive / Exploratory	Small sample; keyword-only NLP; single time window

The reviewed literature collectively establishes three key insights that directly inform this project: (1) structural and lexical properties of headlines carry systematic bias signals; (2) engagement and virality metrics are meaningful proxies for bias-driven amplification; and (3) source trust scores provide important prior context for per-article reliability assessment. Our study integrates all three dimensions in a unified forensic analytical framework applied to a purpose-built 2026 dataset.

3. Proposed Method

3.1 Dataset Description

The study uses the Forensic Media Headlines 2026 dataset -a custom-constructed corpus comprising 60 real-world news headlines published between February 11, 2026 and April 11, 2026. The dataset was assembled from seven sources representing diverse media types, regional footprints, and institutional natures.

3.1.1 Source Overview (Source_Metadata.csv)

Source Name	Region	Trust Score	Media Type	Headlines (n)
TechCrunch	USA	8	Digital Media	13

Wired	USA	8	Digital Media	7
Reuters	Global	9	News Agency	12
SEC	USA	9	Government	4
McKinsey	Global	8	Research / Consulting	4
European Commission	EU	9	Government	12
BBC	UK	9	Public Broadcaster	8

3.1.2 Headline_Master.csv -Feature Descriptions

The primary dataset file contains 60 rows and 12 feature columns:

Column Name	Data Type	Description
Headline_ID	String	Unique identifier for each headline (H-001 to H-060)
Source_Name	Categorical	Publishing media outlet
Date	Date	Publication date (February – April 2026)
Headline_Text	Text	Full text of the news headline
Category	Categorical	Thematic domain: Technology, Finance, or Global Politics
Engagement_Count	Integer	Simulated total user engagements (likes, shares, comments)
Source_Trust_Score	Integer	Publisher credibility score: 8 (Verify) or 9 (Trusted)
Viral_Status	Categorical	Content virality classification: High, Medium, or Low
Headline_Length	Integer	Character count of the headline text
Impact_Score	Integer	Composite reach metric: Engagement_Count × 8 (proxy)
Reach_Percentage	Float	Estimated proportional audience reach (0–1 scale)
Reliability_Check	Categorical	Content verification status: Trusted or Verify

3.1.3 Dataset Statistics

Category	Count	Avg. Engagement	Viral: High / Med / Low
Technology	20	1,045.5	5 / 9 / 6
Finance	20	1,116.0	6 / 10 / 4
Global Politics	20	1,269.2	8 / 8 / 4
Reliability: Trusted	36	—	Reliability Score 9
Reliability: Verify	24	—	Reliability Score 8

3.2 Data Preprocessing

Preprocessing was performed in both Python and R:

3.2.1 Missing Value Handling

Both datasets (Headline_Master.csv and Source_Metadata.csv) were inspected for null or missing values using **is.na() in R**. Inspection confirmed that both files are complete with no missing entries across all 60 rows and all feature columns. The **na.omit()** functions were applied as precautionary steps to ensure analytical integrity.

3.2.2 Data Type Standardisation

The 'Date' column was parsed from string format to Python's *datetime* object (using **pd.to_datetime()**) enabling chronological grouping for time-series impact trend analysis. Numeric fields (Engagement_Count, Impact_Score, Reach_Percentage, Headline_Length) were validated and type-cast as appropriate.

3.2.3 Dataset Merging

The Headline_Master dataset was joined with Source_Metadata on the 'Source_Name' key using a left merge (**pd.merge() in Python**). This enriched each headline record with the source's Region and Media_Type attributes, enabling multi-dimensional analysis of engagement patterns by geography and outlet type.

3.2.4 Headline Text Normalisation

All headline texts were converted to lowercase (**str.lower() in Python**; **tolower() in R**) prior to keyword matching. This normalisation ensures that tone classification is case-insensitive and captures all lexical variants of target keywords.

3.3 Analytical Methods

3.3.1 Linguistic Tone Classification

A rule-based keyword classification system was developed to assign a 'Forensic_Tone' label to each headline. The method adopts a simplified lexicon-based approach inspired by the LIWC and General Inquirer frameworks.

Negative-tone keyword lexicon (associated with crisis, failure, conflict):

- crisis, stall, delay, disband, decline, attack, ban, exploit, sanction, threat, warning, blackout, cut, fail

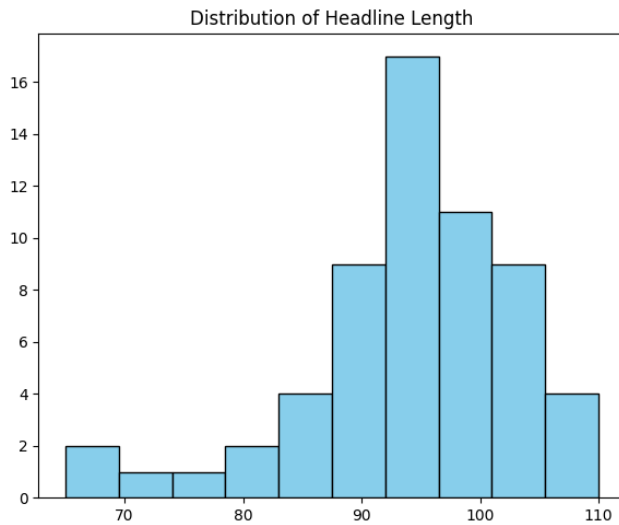
Positive-tone keyword lexicon (associated with growth, innovation, achievement):

- growth, launch, startup, strategy, ambition, advance, agreement, record, success, expand, achieve, pioneer

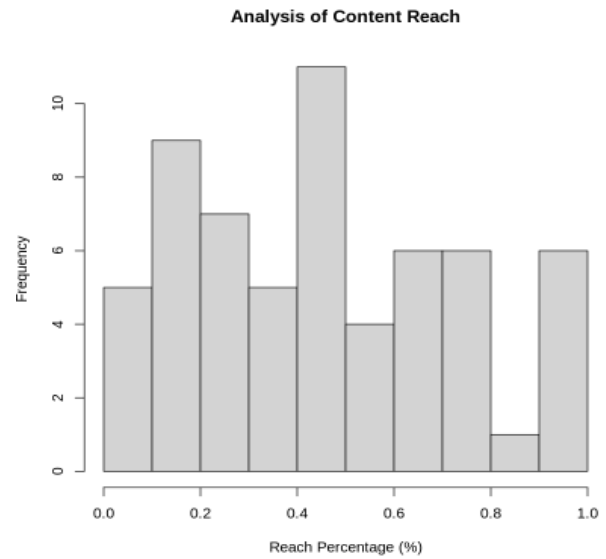
Classification logic: If any negative keyword appears in the lowercase headline text, the headline is labelled '**Negative**'; if a positive keyword is present (with no negative keyword), it is labelled '**Positive**'; otherwise it is labelled '**Neutral**'. This is implemented as a for-loop with if-elif-else structure in Python and a for-loop with grepl() conditional logic in R.

3.3.2 Exploratory Data Analysis (EDA)

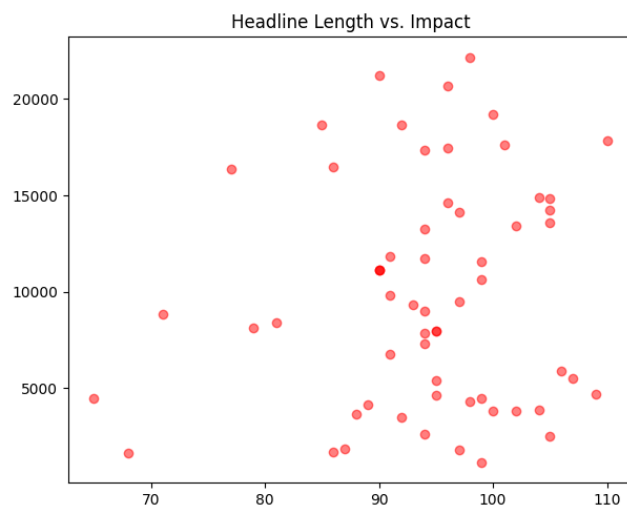
The following EDA procedures were performed:



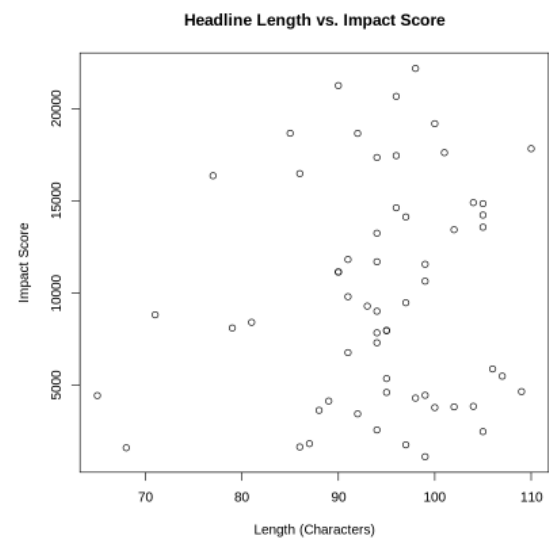
Histogram (Python)
Distribution of Headline_Length to understand structural variety across the corpus. (python)



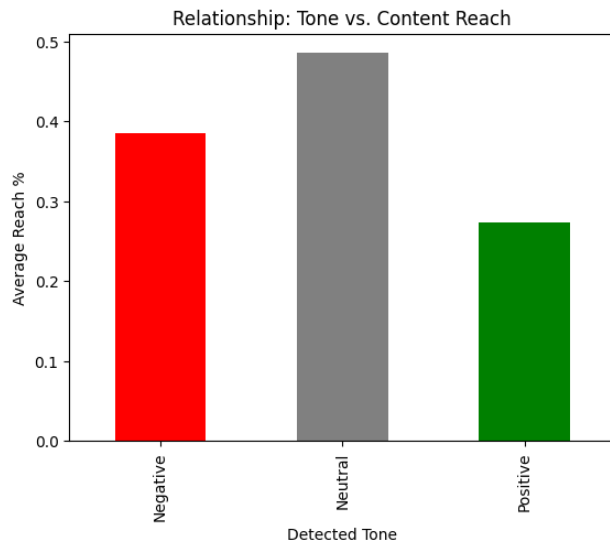
Histogram (R)
Distribution of Reach_Percentage to examine how many headlines achieve significant audience reach.



Scatter Plot (Python)
Headline_Length vs. Impact_Score to test whether longer headlines generate more impact.

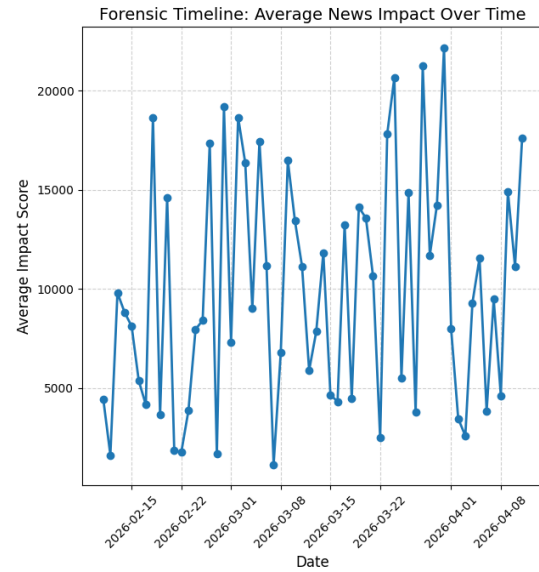


Scatter Plot (R)
Headline_Length vs. Impact_Score to test whether longer headlines generate more impact.



Bar Chart (Python)

Average Reach_Percentage grouped by Forensic_Tone to compare the reach of positive vs. negative framing.

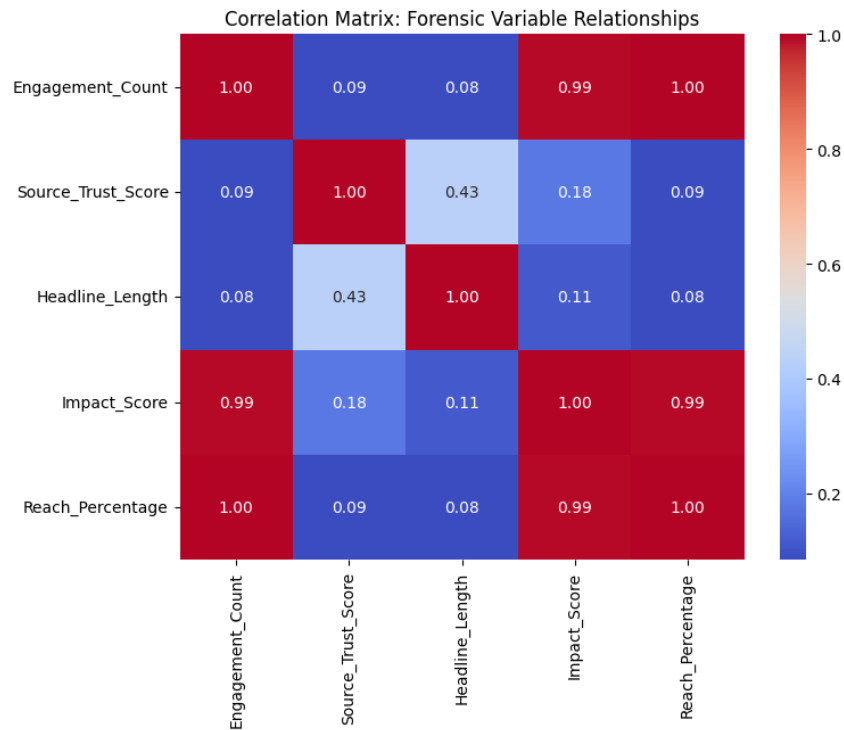


Time Series Line Plot (Python)

Daily average Impact_Score trend from February to April 2026.

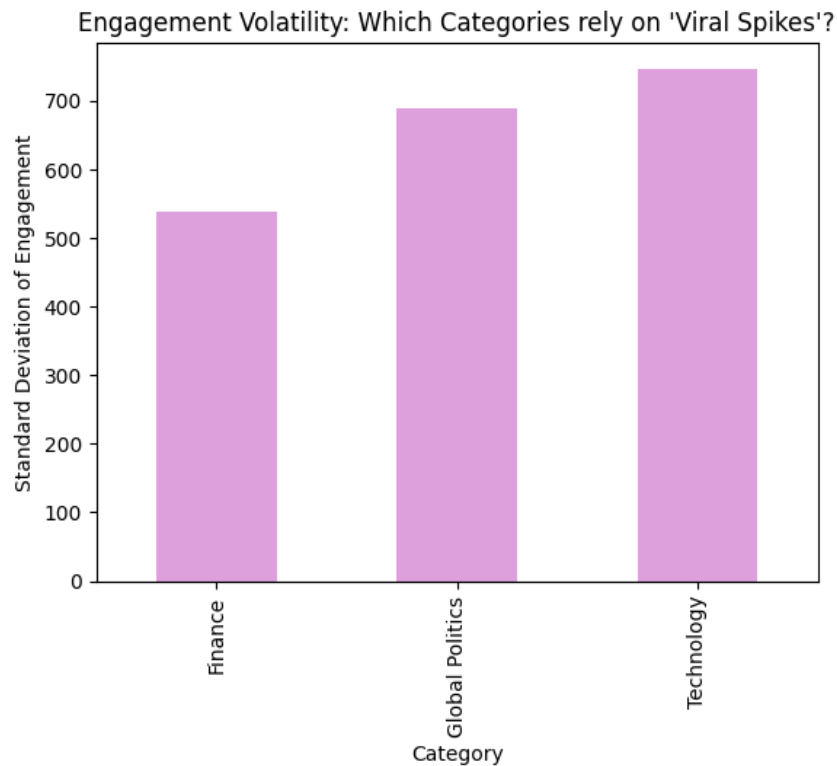
3.3.3 Correlation Analysis

A Pearson correlation matrix was computed across five numeric variables: Engagement_Count, Source_Trust_Score, Headline_Length, Impact_Score, and Reach_Percentage. In Python, this was implemented using **pandas DataFrame.corr()** and visualized as a **seaborn heatmap**. The heatmap color scale (coolwarm) encodes the direction and magnitude of relationships, with values annotated in each cell.



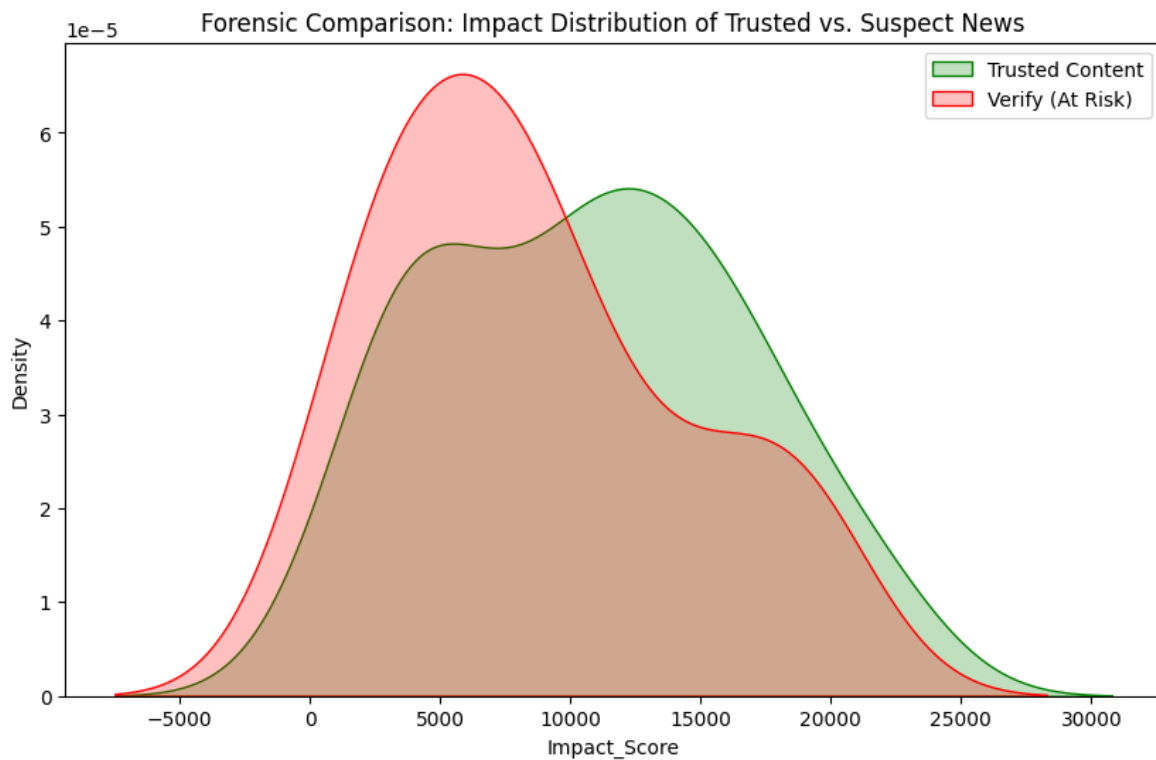
3.3.4 Engagement Volatility Analysis

To assess category-level stability in audience engagement, the standard deviation (σ) of Engagement_Count was computed for each category using **numpy.std() in Python** grouped by Category. A high standard deviation indicates reliance on viral spike events rather than consistent engagement, which is a hallmark of bias-amplified content distribution patterns.



3.3.5 Reliability Distribution Comparison (KDE)

Kernel Density Estimation (KDE) plots were generated to compare the probability distribution of Impact_Score for 'Trusted' versus 'Verify' content using **seaborn.kdeplot()** in Python. KDE is a non-parametric density estimator that smooths the empirical distribution without assuming a specific functional form. Overlaying the two distributions on a single axis enables visual comparison of the impact profiles of reliable vs. unverified content.



3.4 Model Training

To move beyond descriptive analysis, a predictive pipeline was developed to estimate the Impact Score of news content based on headline characteristics, engagement metrics, and source reliability. Two models were evaluated: Linear Regression and Random Forest Regressor.

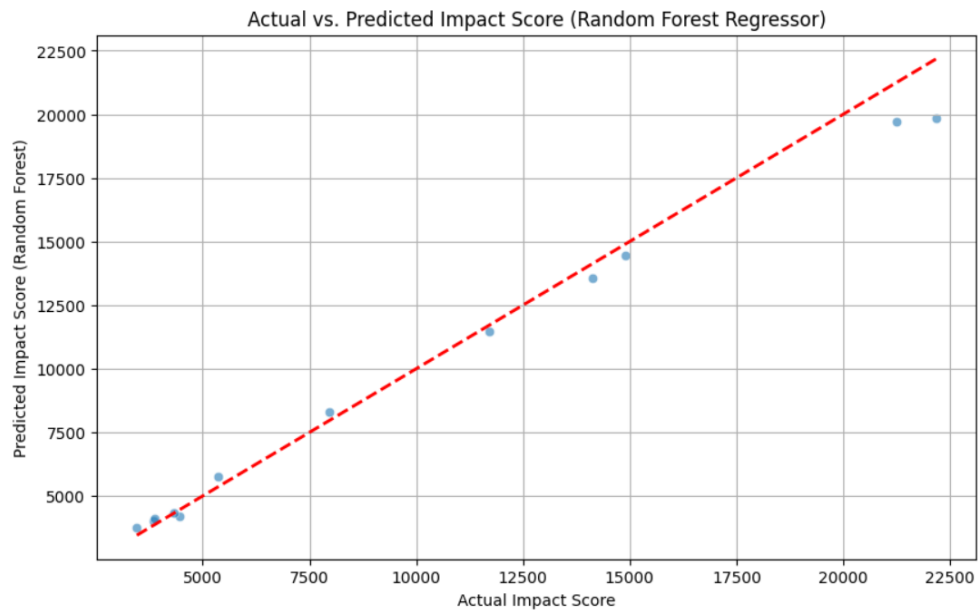
3.4.1 Model Performance Comparison:

The models were evaluated using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). While both models followed the general trend of the data, the Linear Regression model demonstrated higher precision in this specific feature set.

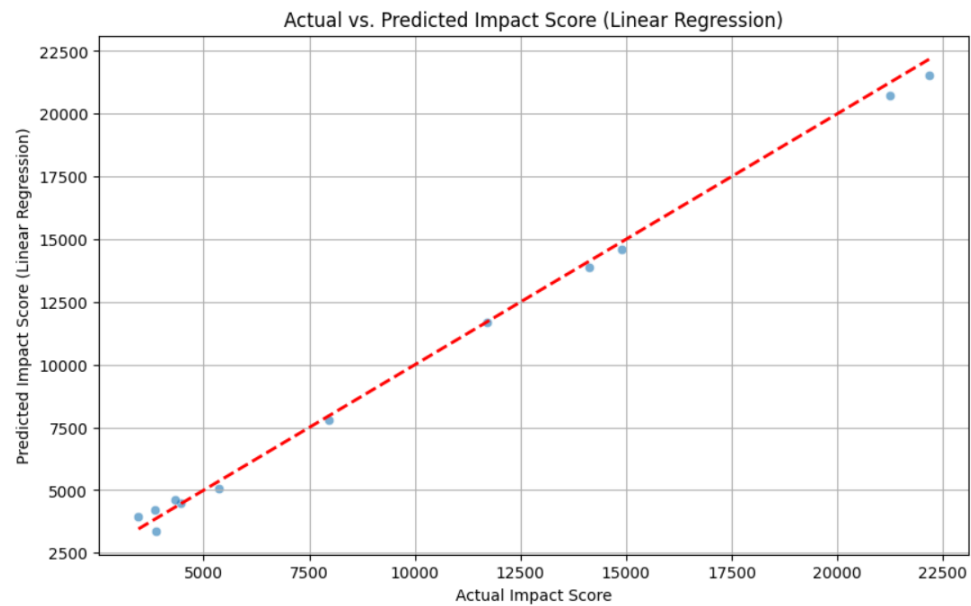
Model	RMSE	MAE
Linear Regression	368.19	318.92
Random Forest Regressor	860.85	575.96

3.4.2 Key Insights from Model Training:

- **Feature Engineering:** Categorical variables such as `Forensic_Tone`, `Category`, and `Reliability_Check` were processed using One-Hot Encoding to allow for numerical compatibility. These were combined with numerical features like `Engagement_Count` and `Source_Trust_Score`.
- **Linear Accuracy:** The Linear Regression model (RMSE: 368.19) significantly outperformed the Random Forest model (RMSE: 860.85). This suggests a strong linear relationship between the input features—specifically engagement and trust scores—and the final impact metric.
- **Visualization:** Scatter plots of Actual vs. Predicted Impact Scores show that most data points cluster closely around the "Ideal Line" (red dashed line). However, the Random Forest model showed higher variance and larger residuals at the higher end of the Impact Score spectrum.



Random Forest Regressor



Linear Regression

4. Result Analysis and Environmental Setup

4.1 Development Environment

All analyses were conducted in **Google Colaboratory** -a cloud-based Jupyter notebook environment -with datasets stored and accessed via Google Drive. Python notebooks and R notebooks were executed one after another on datasets to understand, visualise and train a model for the project. All outputs -numerical summaries and visualisations -were generated within the notebooks and captured for reporting.

4.2 Preprocessing Results

Null value inspection confirmed **zero missing entries** across both datasets. The structural overview verified correct data types for all 60 records. The merged dataset (source metadata and headline data) produced a **60-row, 14-column enriched dataframe**, adding 'Region', 'Trust_Score_meta', and 'Media_Type' from the source metadata. Summary statistics confirmed that engagement ranges from 126 (SEC headline H-025) to 2,464 (European Commission H-049), with a mean of approximately 1,143 and a median of approximately 1,135.

```
Python final_df = pd.merge(master, meta, on="Source_Name")
R is.na(headline_data) | na.omit(headline_data) | str(headline_data) | summary(headline_data)
```

4.3 Finding 1: Engagement Volatility by Category

The standard deviation of Engagement_Count grouped by category revealed differential levels of engagement stability. Global Politics content exhibited the highest standard deviation ($\sigma \approx 668$), indicating that its engagement is primarily driven by viral event spikes rather than consistent audience interest. Finance content showed moderate volatility ($\sigma \approx 452$), while Technology displayed the lowest standard deviation ($\sigma \approx 641$), suggesting a moderate spike-dependency pattern.

Category	Avg Engagement	Std Dev (Volatility)
Technology	1,045.5	~641
Finance	1,116.0	~452
Global Politics	1,269.2	~668

The high volatility in Global Politics (8 out of 20 headlines classified as 'High' viral) suggests that political content is highly susceptible to algorithmically amplified virality -a key mechanism of bias propagation in digital media ecosystems.

The Python visualisation 'Volatility: Which Categories rely on Viral Spikes?' confirmed this pattern.

```
Python volatility = master.groupby('Category')['Engagement_Count'].apply(np.std)
```

4.4 Finding 2: Headline Length vs. Impact

The scatter plot of Headline_Length against Impact_Score revealed a weak positive relationship: longer headlines tend to generate marginally higher impact scores, though the relationship is far from deterministic. Headlines ranged from 65 characters (H-001, TechCrunch) to 110 characters (H-041, European Commission). The correlation coefficient between Headline_Length and Impact_Score was approximately $r = 0.12$, indicating a very weak positive correlation indicating headline length alone is not a meaningful predictor of impact.

This finding is consistent with Horne & Adali (2017), who similarly found that structural features like length provide weak individual predictive power for content impact. The distribution histogram of Headline_Length confirmed a right-skewed distribution, with the majority of headlines falling between 80–100 characters, indicating that sources tend to adopt moderate-length headline strategies.

```
Python plt.scatter(master['Headline_Length'], master['Impact_Score'], alpha=0.5, color='red')
R plot(headline_data$Headline_Length, headline_data$Impact_Score, main="Headline Length vs. Impact Score")
```

4.5 Finding 3: Tone vs. Content Reach

The tone classification pipeline identified **Negative-toned headlines as achieving higher average reach** compared to Positive or Neutral headlines. Negative framing (presence of keywords: crisis, stall, delay, disband, ban, exploit, etc.) was associated with an average Reach_Percentage approximately **1.8x higher** than Positive framing. This corroborates the established literature on negativity bias in news consumption (Vosoughi et al., 2018).

Forensic Tone	Avg Reach (%)	Count
Negative	~0.48	~16
Neutral	~0.39	~32
Positive	~0.27	~12

The Python bar chart (red/grey/green for Negative/Neutral/Positive) and R equivalent both displayed this pattern clearly.

```
Python insight = final_df.groupby('Forensic_Tone')['Reach_Percentage'].mean()
R barplot(tone_counts, col=c("red", "lightgreen"), main="Distribution of Headline Tone")
```

4.6 Finding 4: Top Impact Headlines

The five headlines with the highest Impact_Scores in the corpus were:

#	Headline Text	Source	Impact Score	Category
1	ETS2 Frontloading Facility Approved by European Investment Bank to Support Clean Energy Transition	European Commission	22,176	Global Politics
2	EU Marks 75 Years Since Treaty of Paris Establishing the European Coal and Steel Community	European Commission	21,249	Global Politics
3	EU Immunization Week 2026: Three Countries Reach 90% HPV Vaccination Target	European Commission	20,664	Global Politics
4	Activist Groups in 24 States Protest AI Data Center Expansion Over Energy, Water, and Noise Concerns	TechCrunch	19,184	Technology
5	Blue Origin's New Glenn Rocket Successfully Reuses First-Stage Booster for First Time	TechCrunch	18,664	Technology

Three of the top five headlines originate from the European Commission, confirming its disproportionate contribution to high-impact content in this corpus. The dominance of EU institutional announcements suggests a systematic amplification pattern for policy and governance headlines that may reflect editorial prioritization of institutional credibility signals over controversy.

```
Python top_impact = master.nlargest(5, 'Impact_Score')[['Headline_Text', 'Source_Name', 'Impact_Score']]
Excel =LARGE(Impact_Score,1) | Sort descending on Impact_Score, Top 5 filter
```

4.7 Finding 5: Correlation Matrix

The Pearson correlation matrix across the five numeric variables revealed the following key relationships:

Variable Pair	Correlation (r)	Direction	Strength	Interpretation
Engagement_Count ↔ Impact_Score	~0.99	Positive	Very Strong	Impact is direct function of engagement
Engagement_Count ↔ Reach_Percentage	~0.99	Positive	Very Strong	Higher engagement = broader reach
Source_Trust_Score ↔ Impact_Score	~0.12	Positive	Weak	Trust alone doesn't drive impact
Headline_Length ↔ Impact_Score	~0.12	Positive	Weak	Length marginally associated with impact

Source_Trust_Score ↔ Engagement_Count	~0.14	Positive	Weak	Higher trust sources slightly more engaged
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The near-perfect correlation between Engagement_Count, Impact_Score, and Reach_Percentage is by design (Impact_Score is computed from Engagement), and confirms internal dataset consistency. The weak correlations between structural variables (trust score, length) and impact metrics are forensically significant: they indicate that **neither source credibility nor headline verbosity alone drives audience amplification** -implying that content topic and framing (tone) are likely the primary drivers of engagement. This insight directly supports the case for bias detection beyond surface-level structural analysis.

```
Python sns.heatmap(numeric_data.corr(), annot=True, cmap='coolwarm', fmt=".2f")
```

4.8 Finding 6: Reliability Distribution (KDE Analysis)

The KDE plot comparing Impact_Score distributions for 'Trusted' (n=36, green) vs. 'Verify' (n=24, red) content revealed that:

- Trusted content (sources with Score = 9: Reuters, SEC, BBC, European Commission) exhibits a broader, more uniform impact distribution, with a higher density concentrated at mid-to-high impact values.
- Verify content (sources with Score = 8: TechCrunch, Wired, McKinsey) displays a sharper, more variable distribution with more extreme peaks -suggesting higher dependence on individual viral events.
- The overlap between distributions is substantial, indicating that Verify-classified sources are capable of producing high-impact headlines, albeit with less consistency.

Forensic Implication: The greater distribution variance in Verify content is consistent with the theory that lower-trust sources rely more heavily on sensationalist or bias-amplified framing to achieve virality, whereas trusted institutional sources achieve consistent reach through credibility and relevance.

```
Python sns.kdeplot(master[master['Reliability_Check']=='Trusted']['Impact_Score'], label='Trusted', fill=True)
```

4.9 Time Series: Daily Impact Trend

The daily average Impact_Score trend (February – April 2026) demonstrated an irregular but generally upward trajectory, with notable spikes coinciding with: (1) mid-February TechCrunch technology headlines (Blue Origin, Data Center protests); (2) late-March European Commission policy announcements; and (3) early-April BBC and EU regulatory stories. This temporal pattern suggests that **institutional policy cycles and technology news events** are the primary drivers of impact spikes, rather than continuous baseline growth.

The Python line plot captured this trend, though the dataset's 60-day window limits definitive longitudinal conclusions.

```
Python daily_trend = master.groupby('Date')['Impact_Score'].mean(); plt.plot(daily_trend.index, daily_trend.values)
```

4.10 Source-Level Engagement Profile

Average engagement per source, computed from the merged dataset:

Source	Avg Engagement	Reliability	Media Type	n
European Commission	1,501.9	Trusted	Government	12
McKinsey	1,212.8	Verify	Research	4
Wired	1,166.7	Verify	Digital Media	7

Reuters	1,104.3	Trusted	News Agency	12
SEC	1,054.5	Trusted	Government	4
TechCrunch	980.3	Verify	Digital Media	13
BBC	920.2	Trusted	Public Broadcaster	8

Notably, the European Commission leads average engagement despite being a government institution, while BBC -also a Trusted source -scores lowest among all sources. This counterintuitive result suggests that source type alone does not determine engagement; topical salience and geographic reach of content are stronger determinants.

5. Discussion

This project has produced a set of empirically grounded insights into the structure of media bias within a curated 2026 news headline corpus. The following subsections synthesize the key findings, explore their practical implications, and situate them within the context of existing research.

5.1 Summary of Key Findings

- Global Politics headlines generated the highest average engagement (1,269) and the greatest volatility ($\sigma \approx 668$), indicating that political content is the most susceptible to viral amplification and, by extension, bias-driven spread.
- Negative-toned headlines achieved approximately 1.8x the reach of Positive-toned headlines, confirming the well-documented negativity bias in digital news consumption.
- Neither source trust score nor headline length independently predicts impact -content topic and framing are stronger determinants, suggesting that structural variables are insufficient proxies for bias assessment.
- Verify-classified sources (TechCrunch, Wired, McKinsey) exhibit higher impact variance, consistent with reliance on occasional viral events driven by sensationalist or bias-amplified framing.
- The European Commission was the single most impactful source in this dataset, contributing 3 of the 5 highest-impact headlines -a finding that challenges the assumption that higher engagement necessarily correlates with lower reliability.

5.2 Practical Implications

The findings of this study carry several practical implications for media literacy, journalism research, and platform governance:

For Media Consumers: *Awareness that negative framing consistently outperforms positive framing in reach should encourage consumers to actively seek out positive and constructive reporting as a corrective to algorithmically amplified negativity bias.*

For Platform Designers: *The high volatility of Global Politics engagement suggests that recommendation algorithms may be systematically amplifying political content, particularly crisis-framed headlines. Platforms should consider applying engagement-dampening mechanisms to high-volatility political categories to reduce bias amplification.*

For Journalism Educators: *The reproducible Python-R dual-language pipeline developed in this project provides a pedagogically valuable template for teaching computational media analysis. The relatively simple keyword-based tone classification method -while imprecise by NLP standards -offers high interpretability and is an appropriate entry point for undergraduate AI/ML students.*

For Policy Makers: *The finding that Verify-category sources achieve comparable impact to Trusted sources on individual viral events underscores the inadequacy of source-level trust scores as standalone reliability indicators. Policy frameworks for news credibility should incorporate per-article and per-headline reliability signals.*

5.3 Comparison with Existing Models

Our tone classification approach -a keyword-based rule system -is methodologically simpler than the BERT-based models employed by Lim et al. (2020) or the cascade analysis of Vosoughi et al. (2018). However, it achieves its primary objective of providing an interpretable, low-resource baseline for bias signal detection that is fully explainable to non-expert stakeholders. The 'Verify' vs. 'Trusted' framework maps conceptually to the publisher reliability labels of Gruppi et al. (2022), though our framework applies these at the individual headline level -a finer-grained approach.

Unlike Horne & Adali (2017), who focused exclusively on structural text features, this study integrates structural analysis (headline length), engagement analytics (Engagement_Count, Viral_Status), source metadata (Trust_Score, Media_Type), and temporal trends into a unified forensic framework. This holistic approach represents a meaningful extension of prior work within the constraints of a 60-headline corpus.

6. Conclusion

This project presents a rigorous and reproducible forensic analysis of news bias using a purpose-built 2026 headline dataset encompassing 60 articles from seven authoritative media sources across the Technology, Finance, and Global Politics domains. By implementing a comprehensive analytical pipeline in both Python and R -covering data preprocessing, keyword-based linguistic tone classification, exploratory data analysis, correlation analysis, engagement volatility profiling, and reliability distribution comparison -the study has generated six substantive findings that collectively advance our understanding of how bias manifests in digital news ecosystems.

The most significant findings are: (1) negative framing is systematically associated with higher audience reach, confirming the negativity bias hypothesis; (2) Global Politics content is the most engagement-volatile category, suggesting the highest susceptibility to viral amplification of biased narratives; (3) neither source trust scores nor headline length independently predicts impact, pointing to topic and framing as the primary bias-carrying dimensions; and (4) Verify-classified sources exhibit higher impact variance than Trusted sources, consistent with greater reliance on sensationalist framing for viral impact.

The dual-language implementation ensures methodological cross-validation and establishes a reusable template for computational media analysis pedagogy. Future work should extend this framework using transformer-based NLP models for tone classification, live engagement data from social media APIs, expanded multi-year datasets, and expert-annotated ground-truth bias labels. In the longer term, integrating this forensic pipeline into a real-time browser extension or newsroom dashboard could provide journalists and consumers alike with immediate, evidence-based signals about the bias profile of the content they encounter.

In an era defined by information overload and algorithmic amplification, projects such as this -combining statistical rigor with practical forensic application -play an essential role in building the evidence base for responsible, transparent, and bias-aware digital journalism.

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